

# Effects of disturbance and site factors on sapling dynamics and species diversity in northern hardwood stands

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## ABSTRACT

The northern hardwoods forest ecosystem of eastern North America provides a wealth of products and services to human society. These forests have been managed under partial harvesting treatments to meet wood production objectives of desired species while maintaining or increasing structural and compositional heterogeneity. Few studies have examined how disturbance and site factors interactions may influence sapling dynamics and species diversity in northern hardwood forests. We used a retrospective approach to examine the impacts of disturbance and site factors on sapling (trees > 130 cm in height and < 10.0 cm dbh) dynamics and species diversity across 37 stands with time since harvest ranging from 6 to 20 years. We assessed American beech (*Fagus grandifolia* Ehrh), sugar maple (*Acer saccharum* Marsh.), red maple (*Acer rubrum* L.) and yellow birch (*Betula alleghaniensis* Britt.) saplings density as functions of disturbance intensity (% basal area cut and overstory tree mortality rate (%)) and site factors. We also examined the effects of disturbance, site factors and relative density of American beech, red maple, sugar maple and yellow birch saplings on species diversity at the sapling layer. The density of American beech saplings decreased significantly with harvest intensity. High harvest intensity significantly reduced American beech sapling density at high depth-to-water sites. However, the densities of red maple and yellow birch saplings increased significantly with increasing harvest intensity. We found that high harvest intensities had less negative effect on sugar maple sapling density compared with American beech. Our results also showed that tree species diversity (Shannon diversity index ( $H'$ )) increased with harvest intensity up to about 75% and then declined. However, depth-to-water (DTW) significantly changed the magnitude of tree species diversity response to harvest intensity. Red maple appeared to have contributed more to species diversity than the other hardwood tree species. Given that sugar maple sapling density was generally higher than American beech at high harvest intensities in this study, management regimes that promote mid-tolerant species in American beech-presence stands may eventually benefit sugar maple more than American beech in these forest types. These strategies may also increase species diversity and improve forest ecosystems functions and services.

## 1. Introduction

The emulation of natural disturbance regimes in forest management has increased global interest in alternatives silvicultural systems to conventional clearcutting (Harvey et al., 2002; Haeussler and Kneeshaw, 2003; Drever et al., 2006; Franklin et al., 2007; Long, 2009). In this paradigm shift in forest management, partial harvesting, defined broadly as incomplete removal of merchantable trees, is the commonly used silvicultural practice in many jurisdictions (Harvey et al., 2002; Yoshida et al., 2006; Thorpe and Thomas, 2007; Prévost and Dumais, 2013; Ashton and Kelty, 2018). The objectives may include increasing structural and compositional heterogeneity (Harvey et al., 2002; Kelty

et al., 2003; D'Amato et al., 2015). Complex and compositional diverse forests have positive benefits for forest ecosystems resilience to future climate conditions and disturbances, as well as the maintenance of ecosystem functions and services (Drever et al., 2006; Hupperts et al., 2018).

In uneven-aged forests, candidate alternatives such as group selection, single-tree selection or a hybrid selection approach where gaps are interspersed within single-tree-selection are the common partial harvesting practices (Leak and Filip, 1977; Burns, 1983; MacLean et al., 2010; Ashton and Kelty, 2018). Partial harvest in uneven-aged forests generally relies on natural rather than artificial regeneration for stand establishment (Nyland, 1998; Ashton and Kelty, 2018). Therefore, the

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need for information to guide regeneration techniques is crucial to the sustainable management of timber resources and the maintenance of species diversity in uneven-aged forests.

The operational implementation of partial harvest treatments results in considerable variation of harvest intensities ranging from light to heavy within a stand and or a landscape (Duguid and Ashton, 2013; Bose et al., 2014; Guay-Picard et al., 2015). Variations in harvesting intensities may have different silvicultural and ecological outcomes. For example, light disturbances and associated small canopy gaps perpetuate the dominance of shade-tolerant species while intense disturbances generally favour the recruitment of intermediate shade-tolerant and shade-intolerant species (Runkle, 1981; McClure and Lee, 1993). According to the intermediate disturbance hypothesis (IDH), species diversity is greatest at the medium intensity of disturbances because opportunities exist for both shade-tolerant and midtolerant species to regenerate (Grime, 1973; Connell, 1978; Molino and Sabatier, 2001; Bongers et al., 2009). However, given that some uneven-aged forests are characterized by a rich understory of shade-tolerant tree species (McClure and Lee, 1993; Arie and Lechowicz, 2002; Nolet et al., 2008), the outcome of overstory disturbance may not follow the IDH. For example, density of shade-tolerant pre-existing saplings can outnumber mid-tolerant species even in the largest gaps (McClure and Lee, 1993). Moreover, competing vegetation, ungulate herbivory, and climate-driven constraints such as drought and high temperature are major impediments to forest regeneration (Côté et al., 2004; Kern et al., 2012; Bataineh et al., 2013; Kern et al., 2013; Williams et al., 2013; Harvey et al., 2016; Bose et al., 2017; Bose et al., 2018; Webster et al., 2018). The effects of these factors make it difficult to predict future regeneration dynamics and tree species diversity (Paluch, 2005; Rodríguez-García et al., 2010; Bataineh et al., 2013).

In the northern hardwood forests of eastern North America, disturbances (i.e. silvicultural treatments and or overstory tree mortality) occur in a range of landforms with wide spatial variability in soil and topographical characteristics (Burgess and Wetzel, 2000; Kelty et al., 2003; MacDonald et al., 2004; Ashton and Kelty, 2018; Hupperts et al., 2018; Webster et al., 2018). Variations in soil and topographic attributes results in stand- or and landscape-scale resource heterogeneity that can affect the establishment of species with different life history strategies (Battaglia et al., 1999; Lee et al., 2005; van Rensen et al., 2015; Wason and Dovciak, 2017), thereby altering regeneration dynamics and species diversity (Ricklefs, 1977; Huston, 1979; Bartels and Chen, 2010). For example, American beech and sugar maple are found on well-drained and fertile sites, while red maple and yellow birch can regenerate across a range of soil conditions (Lee et al., 2005; Ecosystem Classification Working Group, 2007). However, the magnitude to which disturbance intensity and site factors may interact to influence regeneration dynamics and species diversity is less certain (Duguid and Ashton, 2013). In northern hardwood forests, studies reporting how disturbance (i.e., partial harvest and overstory tree mortality) and site factors (i.e., soil texture, cartographic depth-to-water index (DTW, a proxy for soil moisture conditions), elevation, slope, and site productivity) might interact to affect saplings (trees > 130 cm in height and < 10.0 cm dbh) dynamics and tree species diversity at the sapling layer are rare. DTW can indicate topographic location and soil moisture conditions, and has been shown to be a strong predictor of tree species dynamics (Murphy et al., 2009; Murphy et al., 2011). Examining the effects of disturbance and site factors simultaneously may help facilitate the implementation of forest management strategies that could balance the establishment of desired species and species diversity.

In this study, the first objective was to examine how harvesting and overstory tree mortality rate affects sapling densities of four hardwood tree species: American beech (*Fagus grandifolia* Ehrh), sugar maple (*Acer saccharum* Marsh.), red maple (*Acer rubrum* L.), and yellow birch (*Betula alleghaniensis* Britt.). These species generally coexist in northern hardwood forests in the study area (Ecosystem Classification Working Group, 2007). American beech is extremely shade tolerant and

regenerates profusely following the creation of small to medium canopy opening (McClure and Lee, 1993; Nyland et al., 2006; Wagner et al., 2010; Giенcke et al., 2014), thereby interfering with the establishment of other species (Nyland et al., 2006). Recent studies have also reported that American beech is expanding in its natural range and increasing in abundance due to climate-induced increases in temperature and precipitation as well as soil acidification (Duchesne and Ouimet, 2009; Bose et al., 2017; Wason and Dovciak, 2017). However, mostly due to defects caused by beech bark disease, American beech is less economically valuable compared to sugar maple, yellow birch and red maple (Houston, 1975; Gavin and Peart, 1993; Nyland et al., 2006). This has huge implications for stand productivity in most northern hardwood forests as infected American beech trees show reduced radial growth but can rapidly sprout from root to occupy the understory (Houston, 1975; Gavin and Peart, 1993; Cale and McNulty, 2018).

We hypothesised that (i) saplings densities will show species-specific response along disturbance gradients, with shade-tolerant species densities decreasing with increasing disturbance intensity while mid-shade tolerant species densities would increase as disturbance intensity increases. However, this pattern would be modified by below-ground resource variability and other site factors. Our second objective was to examine the relationship between disturbance and saplings diversity. We hypothesised that (ii) saplings diversity would be consistent with the intermediate disturbance hypothesis, and (iii) spatial variation in microsite would modify the disturbance-species diversity relationship (sensu Ricklefs, 1977; Huston 1979), with species diversity being lower in drier sites than sites where soil moisture is not limiting.

## 2. Materials and methods

### 2.1. Study area

This study was conducted in the Acadian Forest Region in north-western New Brunswick, Canada (Fig. 1). The climate is cool and humid with annual precipitation ranging from 870 to 1750 mm and mean growing degree days (sum of temperature > 5 °C) of 1532.6 (Environment Canada, 2017). The geological substrates consist of non-calcareous slate and greywacke, Ordovician calcareous and argillaceous metasedimentary strata, and Devonian metasedimentary strata. The forest landscape is generally undulating to rolling terrain, with soils consisting of well-drained loams and sandy loams on glacial till ridges and very poorly drained soils on flat areas between low-profile ridges (Ecosystem Classification Working Group, 2007). On the upland site, the vegetation is dominated by shade-tolerant deciduous tree species while coniferous species and mixed-woods dominate the valleys and hill sides, respectively.

The study region, with a unique blend of softwood and hardwood tree species, is one of the most floristically diverse forest ecosystems in eastern North America (Ecosystem Classification Working Group, 2007). Past silvicultural activities have created complex stand structures with abundance of natural regeneration in the area (Hennigar et al., 2016). The dominant hardwood tree species in the study area are American beech, sugar maple, and yellow birch. Other hardwood tree species that occur in lower proportions are red maple, white birch (*Betula papyrifera* Marsh.) and poplar (*Populus* spp.). The common softwood tree species include balsam fir (*Abies balsamea* (L.) Mill.), black spruce (*Picea mariana* (Mill.) BSP), white spruce (*P. glauca* (Moench) Voss), and red spruce (*P. rubens* Sarg.).

#### 2.1.1. Stand selection and data collection

We selected 37 stands across three ecoregions (sites) of New Brunswick (Ecosystem Classification Working Group, 2007) based on stand type, and time since treatment. The selection criteria were aimed at obtaining a large variety of stand combinations. We restricted the stands to those that received only one known harvest entry to eliminate the compounding effect of repeated harvesting on the results. We

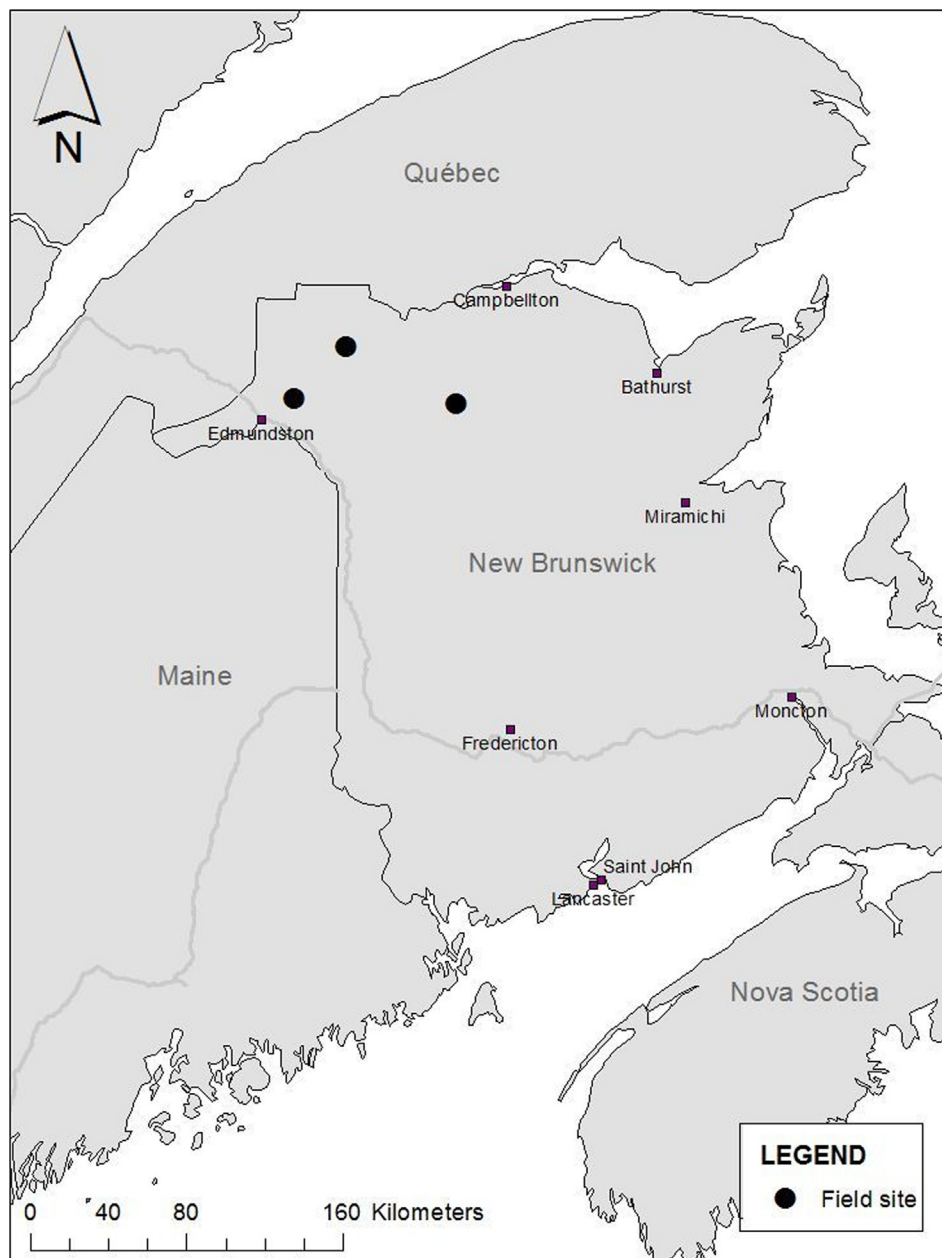


Fig. 1. Location of the study area (black circles) in northwestern New Brunswick.

confirmed the number of harvest entries with harvest records from the New Brunswick Department of Energy and Resource Development (NBDERD) and aerial photography. All the stands had been managed under the broad category of partial harvesting that included single-tree selection, group selection or single-tree with interspersed group cut (Kershaw Jr. et al., 2012). Thus, the study plots consisted of a gradient of harvest intensity (% BA removed), ranging from no harvest (i.e. presence of stump, logging wound, canopy opening) to heavy harvest intensity. The time since harvest of the selected stands ranged from 6 to 20 years.

The sampling unit was a nested plot design consisting of a variable radius plot and a 3.57 m radius (0.004 ha) subplot at the center. The number of plots sampled in each stand ranged from 5 to 9 depending on the size of the stand. Live overstory trees (trees  $\geq 10$  cm diameter at breast height) and snags  $\geq 10.0$  cm diameter at breast height (dbh, 1.3 m above ground) and stumps were sampled in the variable radius plots using a 3 m<sup>2</sup>/ha basal area factor (3BAF) prism from the plot center. This enabled us to quantify the residual basal area (RBA) of

overstory trees and the BA removed during the last harvest. In the 3.57 m radius subplots, all saplings (trees  $> 130$  cm in height and  $< 10.0$  cm dbh) were tallied by species. The dbh was recorded in 2 cm class: 0–1.9 cm; 2–3.9 cm; 4–5.9 cm; 6–7.9 cm; and 8–9.9 cm.

We extracted increment cores at breast height from live overstory trees in four dbh classes (Baral et al., 2016). All increment cores were prepared following standard dendrochronological procedures (Stokes and Smiley, 1996). Annual ring widths were measured to the nearest 0.01 mm using a Velmex UnSlide (Velmex Computer Systems, UK). Cross-dating of each tree ring was done using the package *dplR* (Bunn, 2010) in R (R Core Team, 2018).

#### 2.1.2. Structural diversity

We evaluated tree species diversity at the sapling layer using Shannon diversity index, calculated as

$$H' = - \sum_i p_i \log_2 p_i \quad (1)$$

**Table 1**

Effects of time since harvest (TSH), disturbance (BACUT and MR) and site factors (depth-to-water (DTW) and soil texture (ST)) on hardwoods sapling (tree > 130 cm tall and < 10.0 cm dbh) density in northern hardwood stands in New Brunswick, Canada. Model-averaged estimates, unconditional standard errors (SE) and 95% unconditional confidence intervals (lower and upper) were obtained by multimodel inference.

| Variable              | Model-averaged estimate | Unconditional SE | 95% Unconditional confidence interval |         |
|-----------------------|-------------------------|------------------|---------------------------------------|---------|
|                       |                         |                  | Lower                                 | Upper   |
| <i>American beech</i> |                         |                  |                                       |         |
| Intercept             | <b>6.9574</b>           | 1.0060           | 4.9856                                | 8.9291  |
| TSH                   | −0.0583                 | 0.0597           | −0.1752                               | 0.0587  |
| BACUT                 | − <b>0.0129</b>         | 0.0046           | −0.0220                               | −0.0038 |
| MR                    | 0.0612                  | 0.1226           | −0.1790                               | 0.3013  |
| DTW                   | <b>0.0199</b>           | 0.0095           | 0.0013                                | 0.0385  |
| ST (M-C)              | − <b>1.6995</b>         | 0.5677           | −2.8122                               | −0.5867 |
| BACUT*MR              | − <b>0.0078</b>         | 0.0039           | −0.0154                               | −0.0002 |
| BACUT*DTW             | <b>0.0003</b>           | 0.0002           | 0.0000                                | 0.0006  |
| MR*DTW                | 0.0126                  | 0.0124           | −0.0117                               | 0.0370  |
| <i>Red maple</i>      |                         |                  |                                       |         |
| Intercept             | <b>4.7851</b>           | 1.0570           | 2.7136                                | 6.8567  |
| TSH                   | −0.0761                 | 0.0695           | −0.2122                               | 0.0601  |
| BACUT                 | <b>0.0177</b>           | 0.0048           | 0.0082                                | 0.0271  |
| DTW                   | −0.0003                 | 0.0093           | −0.0186                               | 0.0180  |
| MR                    | 0.1680                  | 0.3344           | −0.4874                               | 0.8233  |
| ST (M-C)              | −0.7106                 | 0.5982           | −1.8831                               | 0.4619  |
| BACUT*MR              | −0.0101                 | 0.0081           | −0.0260                               | 0.0056  |
| BACUT*DTW             | 0.0002                  | 0.0002           | −0.0003                               | 0.0006  |
| MR*DTW                | 0.1836                  | 0.1014           | −0.0151                               | 0.3822  |
| <i>Sugar maple</i>    |                         |                  |                                       |         |
| Intercept             | <b>3.8696</b>           | 0.3235           | 3.2355                                | 4.5037  |
| TSH                   | 0.0399                  | 0.0222           | −0.0035                               | 0.0835  |
| BACUT                 | <b>0.0424</b>           | 0.0071           | 0.0285                                | 0.0562  |
| BACUT <sup>2</sup>    | − <b>0.0004</b>         | 0.0001           | −0.0006                               | −0.0003 |
| MR                    | <b>0.5368</b>           | 0.1339           | 0.2743                                | 0.7993  |
| DTW                   | <b>0.0553</b>           | 0.0092           | 0.0373                                | 0.0733  |
| DTW <sup>2</sup>      | − <b>0.0006</b>         | 0.0001           | −0.0008                               | −0.0004 |
| ST (M-C)              | − <b>0.3983</b>         | 0.2000           | −0.7903                               | −0.0063 |
| BACUT*MR              | 0.0054                  | 0.0029           | −0.0003                               | 0.0110  |
| BACUT*DTW             | −0.0001                 | 0.0001           | −0.0003                               | 0.00004 |
| MR*DTW                | − <b>0.0327</b>         | 0.0096           | −0.0515                               | −0.0138 |
| <i>Yellow birch</i>   |                         |                  |                                       |         |
| Intercept             | <b>5.2733</b>           | 0.6330           | 4.0326                                | 6.5139  |
| TSH                   | 0.0003                  | 0.0445           | −0.0868                               | 0.0874  |
| BACUT                 | − <b>0.0375</b>         | 0.0132           | −0.0634                               | −0.0116 |
| BACUT <sup>2</sup>    | <b>0.0007</b>           | 0.0002           | 0.0003                                | 0.0010  |
| MR                    | <b>0.6655</b>           | 0.2677           | 0.1408                                | 1.1903  |
| BACUT*MR              | − <b>0.0152</b>         | 0.0044           | −0.0238                               | −0.0065 |

Note: BACUT = basal area cut (% basal area removed), BACUT<sup>2</sup> = second order polynomial term, MR = overstory tree mortality rate (%), TSH = time (years) since harvest,

\* Indicates interaction between two variables. Significant ( $p < 0.05$ ) averaged coefficients are highlighted in bold.

where  $p_i$  is the relative abundance of species  $i$ . We calculated this index using the *vegan* package (Oksanen et al., 2018) in R (R Core Team, 2018).

### 2.1.3. Quantifying disturbance

We reconstructed the dbh of all live trees at the time of harvest using the procedure described in Bakker (Bakker, 2005). We obtained the basal area cut using regional species-specific stump-to-breast height equations to estimate dbh of stumps (Wharton, 1984). The basal area of dead trees  $\geq 10$  cm was estimated using the snag decay class information. For detailed information on the reconstruction procedure, see Danyagri et al. (2017).

We estimated harvest intensity in each plot using (Deal and Tappeiner, 2002): (i) the BA of trees  $\geq 10$  cm dbh that survived the harvesting event (living trees at time of harvesting that were still alive

at the time of sampling); (ii) the total BA cut at the harvesting event; and (iii) total mortality BA (BA of trees  $\geq 10$  cm dbh at time of the harvest but died before the sampling date). We also calculated the mortality rate for overstory trees at the plot level using the relationship (Sheil et al., 1995):

$$MR = 100 \left[ 1 - \left( \frac{N_t}{N_0} \right)^{\frac{1}{t}} \right] \quad (2)$$

where

MR = mortality rate (%) of overstory trees in each plot

$N_0$  = number of trees  $\geq 10$  cm alive immediately after harvest in the plot

$N_t$  = number of live-trees immediately after harvest that were still alive at the time of sampling in the plot

$t$  = time since harvest.

### 2.2. Site factors

We obtained a suite site factors to explain the patterns of saplings dynamics and species diversity. The site factors included cartographic depth-to-water (DTW, m), biomass growth index (BGI;  $\text{kg}^{-1} \text{ha}^{-1} \text{yr}^{-1}$ ), slope ( $^\circ$ ), and soil texture (ST). BGI is a quantitative and species-independent measure of site productivity for the Acadian Forest Region (Hennigar et al., 2016). DTW, slope and soil texture were extracted from digital elevation models (DEMs) for New Brunswick using ArcGIS (v. 10.3). BGI was extracted from biomass growth index map for the Acadian Forest Region of New Brunswick.

### 2.3. Statistical analyses

#### 2.3.1. Modeling sapling density

We fitted a generalized linear mixed-effect models (GLMM) with negative binomial error distribution and a log link function in the ‘glmer.nb’ function in the *lme4* package (Bates et al., 2015) to assess the effects of disturbance and site factors on sapling density. The fixed effects included time since harvest (TSH), harvest intensity (% overstory basal area removed, BACUT), mortality rate (%MR) and the site factors (i.e., DTW and ST) presented above. The continuous variables were either included as a linear or second order polynomial terms. The random effects consisted of plot nested within stand, and stand nested within site. We followed the protocol outlined by Zuur et al. (2010) to identify outliers and assess collinearity among predictor variables. We used the ‘evif.mer’ function in the *car* package (Fox and Wisberg, 2011) to detect multicollinearity between predictor variables. Predictor variables with a variance inflation factor (vif) > 2 were removed.

For each tree species (American beech, red maple, sugar maple and yellow birch), we developed ten candidate models with sapling density as the response variable and the disturbance and site factors as predictor variables (Table S1). The form of the disturbance and site factors in the candidate models for each tree species were based on the species’ autecological characteristics. To select the best model, the candidate models were ranked according to the Akaike’s Information Criterion (AIC) corrected for small sample size ( $AIC_c$ ) using the function ‘aictab’ in the *AICcmodavg* package (Mazerolle, 2017). We considered the top-ranked model as one with  $AIC_c$  weight ( $\omega_i$ )  $\geq 0.90$  (Burnham and Anderson, 2002). When the top-ranked model had an  $\omega_i < 0.90$ , a multimodel inference (model averaging) was used to obtain model-averaged estimates, unconditional standard errors (SE) and 95% confidence intervals (Table 1) based on the entire model set (Burnham and Anderson, 2002; Mazerolle, 2017). The model-averaging was performed using the ‘modavg’ function in the *AICcmodavg* package (Mazerolle, 2017). The model-averaged estimates were used in prediction. The *pseudo-R-squared* for both the fixed and random effects ( $R^2_c$ ) and the fixed effects only ( $R^2_m$ ), were obtained using the function

“r.squaredGLMM” in the *MuMIn* package (Barton, 2018).

### 2.3.2. Modeling tree species diversity

We used linear mixed-effects model (LMM) to model the relationship between species diversity (Shannon diversity index) and the predictor variables. We used the function *lmer* in the *lme4* package (Bates et al., 2015). We performed two analyses to obtain the best models. Firstly, we developed six candidate models using time since harvest (TSH), disturbance (BACUT and %MR), site factors (DTW and soil texture) and the relative densities of American beech, red maple, sugar maple and yellow birch saplings (Table S2). The models comparison showed that the model with relative density of sapling (model 6) was the best model (Table S2). However, to assess the effects of disturbance and site factors on species diversity, we performed a second analysis using only disturbance and site factors (Table S3). For both set of analyses, we fitted the linear and quadratic terms of all continuous variables to account for nonlinearity. Site and stand were included as random effects. The assumptions of normality and variance homogeneity of the residuals were visually inspected. All the assumptions were met. Multimodel inference for the second analysis was performed using the procedure described above.

All the statistical analyses were performed using R statistical software (R Core Team, 2018).

## 3. Results

### 3.1. Sapling density

The best models for predicting sapling density varied according to species (Tables S1 and S2). For American beech, model 7 had the lowest AICc. However, the  $\omega_i$  was  $< 0.90$ , indicating that other models had strong support. Overall, seven models had a cumulative  $\omega_i \geq 0.90$ . Similar to the top ranked model for American beech, the top-ranked model (model 4) for red maple had an  $\omega_i < 0.90$  (Table S1). For sugar maple, the model which included TSH, disturbance and site factors without interaction (model 7) had the lowest AICc. The support for this model, however, was insufficient ( $\omega_i < 0.90$ ) to be classed as the “best” model. Due to the strong model selection uncertainty for American beech, red maple and sugar maple, multimodel inferences were conducted to obtain the model-averaged estimates of the predictor variables (Table 1). The top-ranked model for yellow birch was model 5, with  $\omega_i \geq 0.90$  (Table S1). This model included TSH and the main and interaction effects of the disturbance factors (Table 1). Therefore, the parameter estimates for this model were used in the prediction.

Among all the predictor variables, harvest intensity had a significantly negative effect on American beech sapling density (Fig. 2). However, the densities of red maple and yellow birch saplings responded positively to harvest intensity (Fig. 2). For sugar maple, the sapling density showed a unimodal relationship with harvest intensity (Fig. 2). Significant interaction between harvest intensity and DTW was observed for American beech sapling density. Generally, beech sapling density increased with DWT, but the trend was dependent on harvest intensity (Fig. 3A). Depth-to-water also had a significant quadratic relationship with sugar maple sapling density. Sugar maple sapling density peaked between 35 and 45 DWT and declined thereafter with increasing DWT (Fig. 3B). Sugar maple sapling density response to DWT was also influenced by overstory tree mortality rate. Sugar maple sapling density was generally higher when overstory tree mortality rate was higher across the range of DTW (Fig. 3B). The response of yellow birch sapling density to overstory tree mortality rate also depended on harvest intensity. Mortality rate of overstory trees had a positive effect on the density of yellow birch sapling at low and intermediate harvest intensity, but a negative effect at high harvest intensity (Fig. 4).

### 3.2. Tree species diversity

The best model describing  $H'$  in the first analysis was model 6 (Table S2). This model included the relative densities of American beech, red maple, sugar maple and yellow birch.  $H'$  showed a hump-shaped trend as the relative densities of the four hardwood tree species increase (Table 2, Fig. 5). Among the species,  $H'$  was generally higher in higher relative density of red maple compared with the other hardwood tree species. In the second analysis, the top-ranked model describing  $H'$  was model 3 with an  $\omega_i = 0.75$  (Table S3). This model included the TSH, the linear and quadratic terms of BACUT, %MR, DTW, and the interaction between the disturbance factors and DTW. However, due to the strong model selection uncertainty (i.e.  $\omega_i < 0.90$ ), multimodel inference based on candidate model set was conducted to obtain the model-averaged estimates of the predictor variables (Table 3).  $H'$  increased at small to medium harvest intensities, reached a plateau between 60 and 80%, and then declined as harvest intensity increased (Fig. 6). The  $H'$  response to harvest intensity varied depending on DTW, increasing from low to high DTW (Fig. 6).

## 4. Discussion

The emulation of natural disturbance processes has caused a shift in forest management approaches focussed exclusively on timber production to a broader set of ecosystem values and functions (Harvey et al., 2002; Long, 2009). In this shift in forest management, partial harvesting is used as a tool under different silvicultural systems to increase structural and compositional heterogeneity in managed stands (Harvey et al., 2002; Kelyt et al., 2003; D'Amato et al., 2015). Regeneration success following harvest plays a key role in determining future stand dynamics and species diversity (Kern et al., 2013; Danyagri et al., 2017) crucial to the sustainable management of timber resources and the maintenance of species diversity in uneven-aged forests. In this study, we examined how disturbance intensity affects American beech, red maple, sugar maple and yellow birch saplings dynamics and tree species diversity in northern hardwood stands of eastern North America. In addition, the influence of site factors on the responses of sapling dynamics and species diversity to disturbance was assessed. We found that harvest intensity had significant effects on sapling density of the four hardwood tree species examined in this study, being negative or positive depending on the species. The interaction between harvest intensity and DTW had negative effect on American beech sapling density. The study also shows that depth-to-water also changed the magnitude of tree species diversity response to harvesting intensity. Our study indicates that harvesting intensity should be based on the soil moisture conditions at the site to ensure the maintenance of tree species diversity in tolerant northern hardwood forests.

### 4.1. Sapling density

Among all the factors examined in this study, harvest intensity was a strong predictor of sapling density and species dynamics. Our results clearly showed that American beech and sugar maple sapling densities were generally higher at the low to intermediate harvest intensities. Our observations correspond well with the species' light requirements and the patterns of dominance along disturbance gradients (McClure and Lee, 1993; Wagner et al., 2010; Kern et al., 2013). The high densities of American beech and sugar maple at the lower end of the disturbance gradient can be explained by several factors. For instance, both species are considered to be late-successional species, generally characterized by extreme shade tolerance and very slow growth rate. These traits enable them to form a rich sapling bank in the understory over a prolonged period of time (Beaudet et al., 1999; Wagner et al., 2010), and offer the advantage of *in-situ* recruitment as canopy gaps become available (Greene et al., 1999).

American beech was generally higher at the harvest intensities

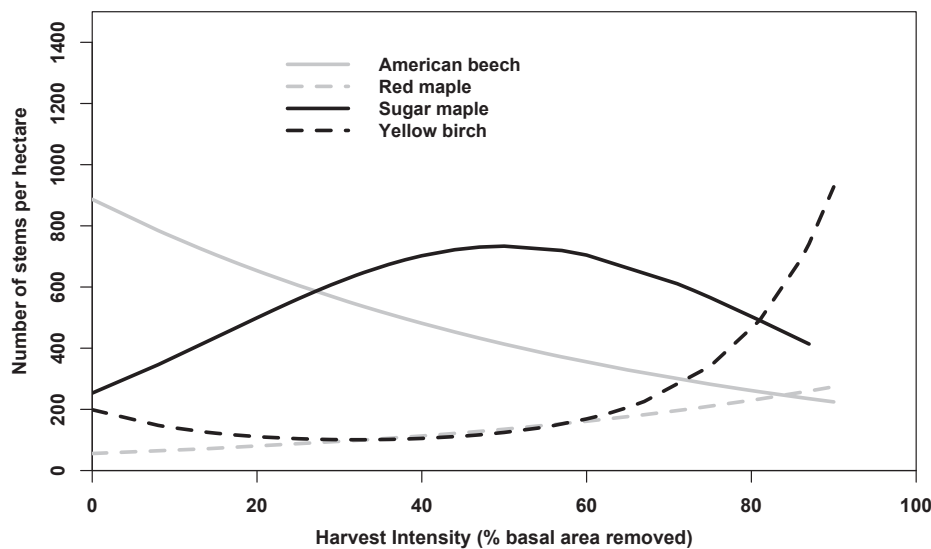


Fig. 2. The effect of harvest intensity (% basal area removed) on the saplings ( $\geq 1.3$  m height and  $< 10$  cm dbh) density of the four hardwood tree species (American beech, red maple, sugar maple and yellow birch) examined in this study.

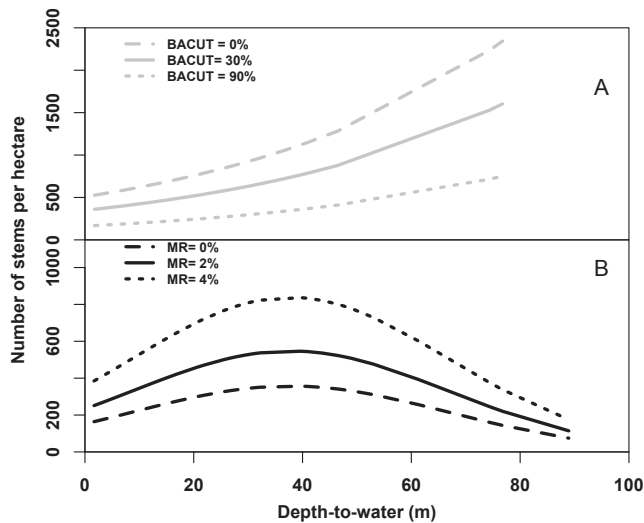


Fig. 3. Effects of the interactions between depth-to-water (DTW) and harvest intensity (% basal area removed) and/or overstory tree mortality rate on American beech (A) and sugar maple (B) saplings densities.

below 30% compared with all the species examined in this study. This disturbance gradient corresponds to single-tree selection harvest (i.e.,  $< 40\%$  basal overstory basal area removal) which favours American beech establishment (Nolet et al., 2008; Wagner et al., 2010; Hupperts et al., 2018). American beech is also capable of vegetative reproduction, and root damage from harvesting activities exacerbates beech root suckering (Beaudet et al., 1999; Beaudet and Messier, 2008; Dracup and MacLean, 2018). Efficient resource utilization under low light environment (Reid and Strain, 1994) has also been implicated as a factor promoting American beech regeneration and abundance across northeastern USA (Bose et al., 2017). In addition, northern hardwood forests are known to support a relatively high population of deer (Burns, 1983). Preferential browsing by deer may have created regeneration opportunities for American beech, an unpalatable browse-species, to increase in abundance (Côté et al., 2004; Matonis et al., 2011; Kern et al., 2012; Bose et al., 2018).

Besides the influence of the species traits and selective browsing of preferred species on the abundance of American beech, climate change-mediated outcomes may also act as triggers for American beech

dominance. In the last decades, increased precipitation and growing season temperature, coupled with an extended growing season, have been reported in northeastern North America (Hayhoe et al., 2007; D'Orangeville et al., 2016; Bose et al., 2017). Both increasing temperature and precipitation, and winter freeze-thaw events have been found to favor American beech suckering and abundance than associated tree species in eastern North America (Jones and Raynal, 1986; Bose et al., 2017; Wason and Dovciak, 2017). On the other hand, climate change-related increases in acid deposition and winter thaw-freeze events have been reported to have negative effects on maples and birches (Auclair, 2005; Bourque et al., 2005; Phillips and Laroque, 2010) that co-exist with American beech. Thus, we speculate that climate-driven factors partly influenced American beech sapling density observed in this study.

One of the challenges of managing tolerant northern hardwood forests in our study area is the proliferation of American beech following canopy disturbance that interfere with the establishment and growth of economically important tree species (Nyland et al., 2006). The close association of American beech and sugar maple, both in terms of their autecological traits (Beaudet et al., 1999; Wagner et al., 2010) and spatial coexistence (Nelson and Wagner, 2014), exacerbates this problem for forest managers. Our results demonstrate that high intensity harvests had a less negative effect on sugar maple compared to American beech. In addition, both red maple and yellow birch saplings densities responded positively to increasing harvest intensity. Red maple and yellow birch are mid-tolerant species that respond favorably to light availability as canopy gap size increases (Eyre and Zillgitt, 1953; Lorimer, 1984; Webster and Lorimer, 2005b). This suggests that high intensity harvests at relatively drier sites in tolerant hardwood forests may provide regeneration opportunities for sugar maple, red maple and yellow birch than American beech. This has potential to diversify the regeneration layer and lead to a more diverse stand in the future. Varying harvest gaps to increase resource heterogeneity has been proposed as a viable model for restoring tree species diversity in eastern North American hardwood forests (Webster et al., 2018). Based on our results, we believe that group selection or hybrid selection approach where gaps are interspersed within single-tree-selection will produce the best results for landowners.

We found that American beech and sugar maple saplings density showed contrasting disturbance-mediated responses to depth-to-water, partly supporting our first hypothesis. At high depth-to-water, high intensity partial harvests reduced American beech sapling density by

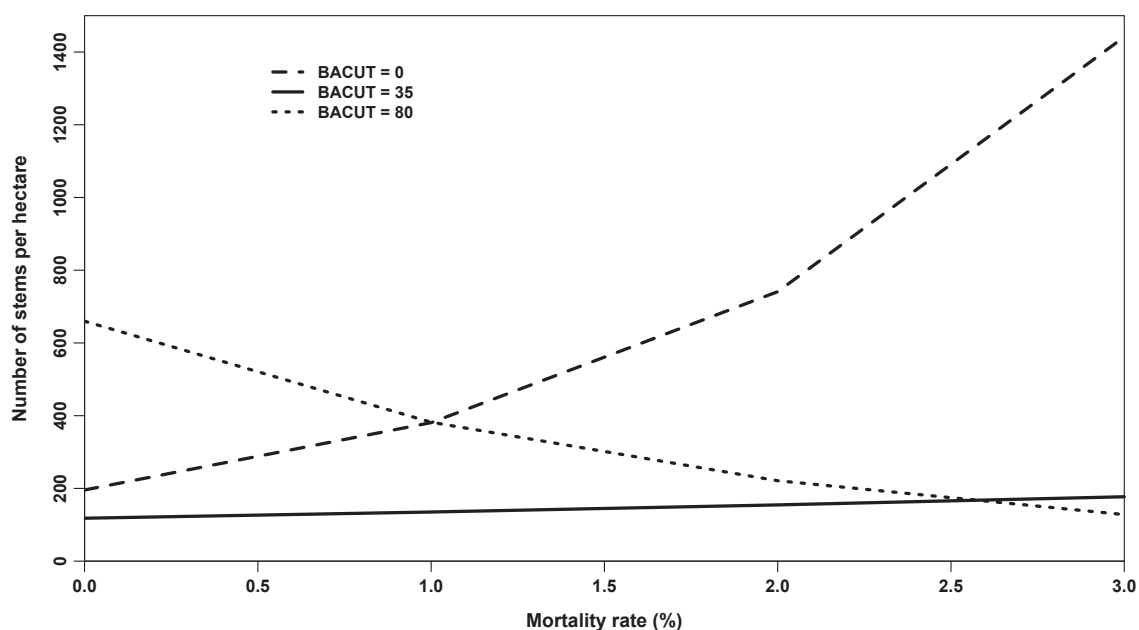


Fig. 4. The effect of the interaction between harvest intensity (% basal area removed, BACUT) and overstory tree mortality rate (%MR) on yellow birch sapling density.

Table 2

Effects of relative densities of American beech (RDBE), red maple (RDRM), sugar maple (RDSM) and yellow birch (RDYB) saplings (trees > 130 cm tall and < 10.0 cm dbh) on species diversity (Shannon diversity index ( $H'$ )) in northern hardwood stands in New Brunswick, Canada.

| Variable          | Model-averaged estimate | Unconditional SE | 95% Unconditional confidence interval |         |
|-------------------|-------------------------|------------------|---------------------------------------|---------|
|                   |                         |                  | Lower                                 | Upper   |
| Intercept         | <b>0.9732</b>           | 0.0588           | 0.8581                                | 1.0884  |
| RDBE              | <b>1.1670</b>           | 0.3277           | 0.5247                                | 1.8094  |
| RDBE <sup>2</sup> | <b>-2.1951</b>          | 0.4184           | -3.0151                               | -1.3750 |
| RDRM              | <b>0.9519</b>           | 0.3164           | 0.3318                                | 1.5719  |
| RDRM <sup>2</sup> | <b>-1.8850</b>          | 0.3796           | -2.6289                               | -1.1411 |
| RDSM              | <b>1.4901</b>           | 0.4979           | 0.5142                                | 2.4660  |
| RDSM <sup>2</sup> | <b>-2.3331</b>          | 0.9887           | -4.2708                               | -0.3953 |
| RDYB              | <b>1.3072</b>           | 0.3332           | 0.6541                                | 1.9603  |
| RDYB <sup>2</sup> | <b>-2.3746</b>          | 0.5435           | -3.4399                               | -1.3094 |

Note: RDBE<sup>2</sup>, RDRM<sup>2</sup>, RDSM<sup>2</sup> and RDYB<sup>2</sup> = second order polynomial terms, SE = standard error. Significant ( $p < 0.05$ ) averaged coefficients are highlighted in bold.

about 80% compared with the low harvest intensity. On the contrary, sugar maple sapling density increased from low to high overstory tree mortality across the range of depth-to-water examined in this study. Our study did not observe any significant effects of depth-to-water on red maple and yellow birch saplings density across the disturbance gradients. Depth-to-water can indicate slope position in the Acadian Forest, with low DTW signifying poorly drained flat terrains or stream steep slopes (Murphy et al., 2009; Murphy et al., 2011; Hennigar et al., 2016). Upslope expansion of American beech has observed in north-eastern United States (Wason and Dovciak, 2017), which suggests that the high depth-to-water values in this study were associated with well-drained slopes or at the crest of relief where American beech thrives best. However, the strong negative effect of increasing harvesting intensity on American beech observed in this study may benefit the establishment of other important tree species such as red maple, sugar maple and yellow birch. This may have important implications as forest managers use partial harvesting as a tool to facilitate the establishment of economically valuable tree species at sites where they would

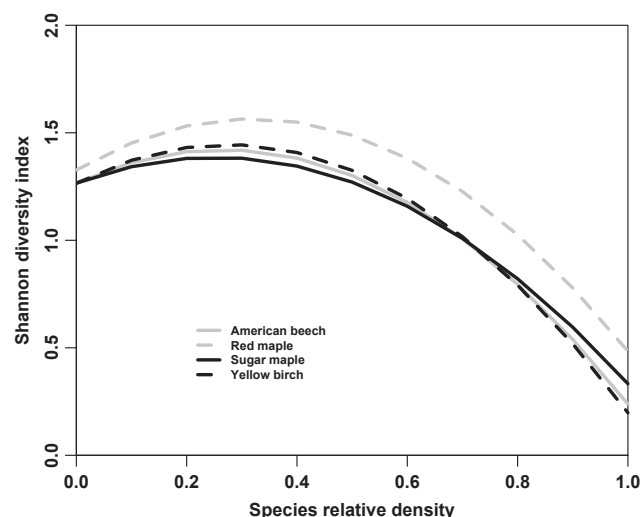


Fig. 5. The relationship between tree species diversity (Shannon diversity index) at the sapling layer and the relative densities of the four hardwood tree species (American beech, red maple, sugar maple and yellow birch) examined in this study.

otherwise be outcompeted.

#### 4.2. Tree species diversity

Our results show that silvicultural treatments and overstory tree mortality rate affect tree species diversity. This demonstrates that these indices are both relevant for understanding the effects of disturbances on tree species diversity at the sapling layer. For harvest intensity, the relationship with  $H'$  was generally unimodal. A possible explanation for the observed pattern of  $H'$  is that the shade-tolerant (e.g. beech and sugar maple) and mid- to intolerant species (e.g., red maple and birches) that occupy both extremes of the harvest intensity range restricted the establishment of other species and subsequently reduced species diversity.

Our study demonstrated that DTW changed the degree to which tree species diversity respond to the gradient of harvest intensity. For a

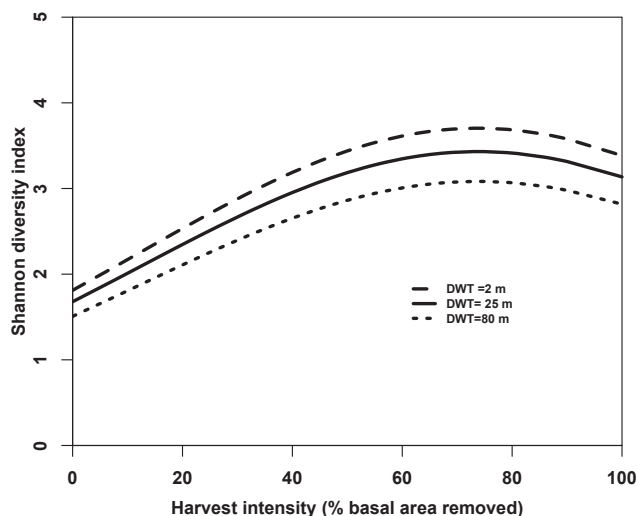
**Table 3**

Effects of time since harvest (TSH), disturbance (BACUT and %MR) and site factors (depth-to-water (DTW) and soil texture (ST)) on species diversity (Shannon diversity index ( $H'$ )) in northern hardwood stands in New Brunswick, Canada. Model-averaged estimates, unconditional standard errors (SE) and 95% unconditional confidence intervals (lower and upper) were obtained by multimodel inference.

| Variable           | Model-averaged estimate | Unconditional SE | 95% Unconditional confidence interval |        |
|--------------------|-------------------------|------------------|---------------------------------------|--------|
|                    |                         |                  | Lower                                 | Upper  |
| Intercept          | <b>0.7109</b>           | 0.1769           | 0.3642                                | 1.0576 |
| TSH                | −0.0087                 | 0.0103           | −0.0288                               | 0.0114 |
| BACUT              | <b>0.0194</b>           | 0.0045           | 0.0106                                | 0.0282 |
| BACUT <sup>2</sup> | <b>−0.0001</b>          | 0.0000           | −0.0002                               | 0.0000 |
| MR                 | −0.0079                 | 0.0465           | −0.0990                               | 0.0833 |
| DTW                | −0.0032                 | 0.0030           | −0.0092                               | 0.0027 |
| ST (M-C)           | −0.0162                 | 0.0953           | −0.2030                               | 0.1706 |
| BACUT*MR           | 0.0001                  | 0.0010           | −0.0018                               | 0.0020 |
| BACUT*DTW          | <b>0.0002</b>           | 0.0001           | 0.0000                                | 0.0003 |
| MR*DTW             | 0.0031                  | 0.0024           | −0.0016                               | 0.0077 |

Note: BACUT<sup>2</sup> = second order polynomial term, ST (M-C) = medium-coarse textured soil.

\* Indicates interaction between two variables. Significant ( $p < 0.05$ ) averaged coefficients are highlighted in bold.



**Fig. 6.** The effect of the interaction between harvest intensity (% basal area removed) and depth-to-water (DTW) on tree species diversity (Shannon diversity index) at the sapling layer.

given level of disturbance, tree species diversity increased more on moisture-rich sites than in the site where soil moisture is limiting. For instance, our study shows that a 25% basal area removal resulted in a Shannon diversity index value of 3 at moist site while 38% and 52% basal area removal are required to obtain the same diversity index at intermediate and relatively drier sites, respectively. Although the magnitude of tree species diversity has reduced in resource poor site (DTW = 80 m) than in resource rich site (DTW = 2 m) in this study, the peak of the tree species diversity curve remained at the same level of disturbance across the gradient of soil moisture regime. This study shows that disturbance and spatial variation in depth-to-water associated with microsite suitability (Ricklefs, 1977; Huston, 1979; Battaglia et al., 1999) are important factors shaping tree species diversity in northern hardwood forests. In a synthesis of studies examining understory species diversity across different forest biomes, Bartels and Chen (2010) also demonstrated that spatial distribution of resources, including soil moisture, was a key driver of species diversity in old-growth forests. This study also shows that there is a tendency for

tree species co-existence in resource rich sites and competitive exclusion to occur in resource poor sites (Nakashizuka, 2001; Chase and Leibold, 2003). The disturbance-sapling densities relationships in this study also show higher density of American beech (superior competitor) sapling at low harvest intensity, while intermediate to high levels of disturbance tended to favor the co-existence of several tree species, such as sugar maple, red maple, yellow birch (Fig. 2). Since sugar maple is more likely to be outcompeted by American beech in well-drained sites such as upslope (Wason and Dovciak, 2017) at lower intensity of harvest (Fig. 3), it is important for forest managers to consider soil moisture regimes when determining the required intensity of harvest to facilitate tree species co-existence and diversity.

Although  $H'$  showed the same trend across all the relative densities of all four hardwood tree species, the relative density of red maple generally had the highest  $H'$ . Red maple is a prolific root sprouters (Cooper-Ellis et al., 1999; Kochenderfer et al., 2004; Plotkin et al., 2013), and behaves both as an early- and a late-successional species (Abrams, 1998). In recent decades, increased expansion of red maple in its native range in eastern north America, partly due to climate change, has been observed (Fei and Steiner, 2007). Our results demonstrate that, while red maple is commercially less desired compared to sugar maple and yellow birch in the study area, this species is important if maintaining species diversity is considered.

## 5. Management implications

Partial harvesting in uneven-aged forests is generally applied to create opportunities for regeneration and to increase species heterogeneity (Kelty et al., 2003; Ashton and Kelty, 2018). Understanding sapling dynamics and tree species diversity relationships to disturbance regimes across varied sites may help forest managers reconcile timber production objectives with biodiversity needs. Our study showed that high harvest intensity reduces the abundance of American beech in upslope (i.e., high DTW sites) where it is considered to be more competitive than sugar maple (Wason and Dovciak, 2017). Higher harvest intensity also benefited red maple and yellow birch, consistent with other findings (McClure and Lee, 1993; Webster and Lorimer, 2005a). The fact that high intensity harvests had less negative effect on sugar maple sapling density compared with American beech in our study suggests that regeneration strategies that target mid-tolerant species may eventually benefit sugar maple establishment and recruitment. Thus, group selection or a combination of single-tree with group selection might be the effective silvicultural systems than single-tree selection alone, especially in stands containing American beech.

The need for compositionally diverse forests is critical in this era of uncertain changes in climate and disturbance events. Compositionally diverse forests are more resilient and adaptable to climate change and disturbance, and can meet present and future resource needs and management objectives (Drever et al., 2006). The challenge for forest managers will be how to balance tree species diversity with timber production. Our results show a strong effect of depth-to-water on tree species diversity response to harvesting intensity. Our results suggest a strong negative species interaction (competitive exclusion) at relatively drier sites along the gradient of harvest intensities. Therefore, effective implementation of partial harvest treatment could benefit from utilizing DTW spatial layers to determine the optimum level of harvest intensity needed to maintain species diversity and recruit desired species in northern hardwood forests.

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## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2019.04.041>.

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